

Lights Out Application built using .Net 8 and Framework Windows Presentation Foundation (WPF) as Windows Desktop application Installer We have initially multiple test cases with multiple depths

Problem #1

Depths: - 2
Boards: - 3 * 3
Cells: - 9
Pieces: - 7

Solution Co-ordinates: - (0,0) (1,0) (1,0) (0,0) (1,0) (0,1)

Solution: -

Problem

Initial Board

1	0	0
1	0	1
0	1	1

Pieces

-	-	X	X
X	X	X	X
X	-	-	X

Piece 1

-	X	X	X
X	X	-	X
-	X	X	X

Piece 2

X	X	X	-
X	X	-	X
-	X	X	X

Piece 3

X	X	X	-
X	X	-	X
-	X	X	X

Piece 4

X	X	X	X
X	X	-	X
X	X	-	X

Piece 5

X	X	X	X
X	X	-	X
X	X	-	X

Piece 6

X	X	X	X
X	X	-	X
X	X	-	X

Piece 7

Solved successfully. Final board is all 0.

Solution coordinates: (0,0) (1,0) (1,0) (0,0) (1,0) (0,1)

Solution

1	0	0
1	0	1
0	1	1

Initial

1	0	1
0	1	0
1	1	1

Piece 1
(0,0)

1	1	1
0	0	0
1	0	1

Piece 2
(1,0)

1	1	0
0	1	1
1	0	1

Piece 3
(1,0)

0	0	0
0	0	1
1	1	0

Piece 4
(0,0)

0	1	1
0	1	1
1	1	0

Piece 5
(1,0)

0	0	0
0	1	1
1	1	0

Piece 6
(1,0)

0	0	0
0	0	0
0	0	0

Piece 7
(0,1)

Problem #2

Depths: - 2
Board: - 4 * 4
Cells: - 16
Pieces: - 7

Solution Co-ordinates: - (1,0) (0,1) (0,1) (2,2) (1,0) (1,0) (0,2)

Solution: -

Sample 2

File: 02.txt
Depths: 2 | Boards: 4 * 4 | Cells: 16 | Pieces: 7

Solve Sample

Problem

Initial Board

0	1	0	0
0	1	1	0
1	0	1	0
1	1	1	0

Pieces

X	X	X	-	-	X	X	X	X
X	X	X	X	X	X	X	X	X
X	X	-	-	-	-	-	-	X

Piece 1

X	X	X	X	X	X	X	X	X
X	X	X	X	X	X	X	X	X
X	X	X	X	X	X	X	X	X

Piece 2

X	X	X	X	X	X	X	X	X
X	X	X	X	X	X	X	X	X
X	X	X	X	X	X	X	X	X

Piece 3

X	X	X	X	X	X	X	X	X
X	X	X	X	X	X	X	X	X
X	X	X	X	X	X	X	X	X

Piece 4

X	X	X	X	X	X	X	X	X
X	X	X	X	X	X	X	X	X
X	X	X	X	X	X	X	X	X

Piece 5

X	X	X	X	X	X	X	X	X
X	X	X	X	X	X	X	X	X
X	X	X	X	X	X	X	X	X

Piece 6

X	X	X	X	X	X	X	X	X
X	X	X	X	X	X	X	X	X
X	X	X	X	X	X	X	X	X

Piece 7

Solved successfully. Final board is all 0.

Solution coordinates: (1,0) (0,1) (0,1) (2,2) (1,0) (1,0) (0,2)

Solution

0	1	0	0
0	1	1	0
1	0	1	0
1	1	1	0

Initial

0	0	0	0
0	1	1	0
1	0	1	0
1	1	1	0

Piece 1
(1,0)

0	0	0	0
0	1	1	0
1	0	1	0
1	1	1	0

Piece 2
(0,1)

0	0	0	0
0	1	1	0
1	0	1	0
1	1	1	0

Piece 3
(0,1)

0	0	0	0
0	1	1	0
1	0	1	0
1	1	1	0

Piece 4
(2,2)

0	0	0	0
0	1	1	0
1	0	1	0
1	1	1	0

Piece 5
(1,0)

0	0	0	0
0	1	1	0
1	0	1	0
1	1	1	0

Piece 6
(1,0)

0	0	0	0
0	1	1	0
1	0	1	0
1	1	1	0

Piece 7
(0,2)

Problem #3

Depths: - 2

*Boards: - 6 * 4*

Cells: - 24

Pieces: - 11

Solution Co-ordinates: - (0,0) (1,3) (0,2) (2,0) (1,0) (0,4) (1,2) (0,0) (0,0) (0,2) (1,0)

Solution: -

Blue Block Lights Out Solver

Creating light out puzzles from the template below, with each tile loaded as a separate, independent puzzle sample.

Sample 1 solved.



Pieces

X	-	-	-
X	-	-	-
X	X	X	-
X	X	X	-

Piece 1

X	X	X	X
X	X	X	-
-	X	-	-
-	X	-	-

Piece 2

X	X	-	-
X	-	X	X
-	X	-	-
-	X	-	-

Piece 3

X	X	-	-
X	-	X	X
-	X	-	-
-	X	-	-

Piece 4

X	X	-	-
X	-	X	X
-	X	-	-
-	X	-	-

Piece 5

-	X	-	-
-	X	-	-
X	X	-	-
X	X	-	-

Piece 6

-	X	-	-
-	X	-	-
X	X	-	-
X	X	-	-

Piece 7

-	X	-	-
-	X	-	-
X	X	-	-
X	X	-	-

Piece 8

X	-	X	-
X	-	X	-
X	-	X	-
X	-	X	-

Piece 9

X	-	X	-
X	-	X	-
X	-	X	-
X	-	X	-

Piece 10

X	-	X	-
X	-	X	-
X	-	X	-
X	-	X	-

Piece 11

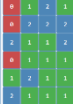
Solved successfully. Final board is all 0.

Solution coordinates: (0,0) (1,2) (0,2) (2,0) (1,0) (0,4) (1,2) (0,0) (0,0) (0,2) (1,0)

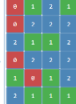
Solution



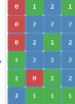
Initial



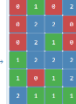
Place 1 (0,0)



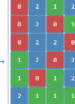
Place 2 (0,2)



Place 3 (1,2)



Place 4 (2,0)



Place 5 (1,0)



Place 6 (0,4)



Place 7 (1,2)



Place 8 (0,0)

Problem #4

Depths: - 3

*Boards: - 5 * 6*

Cells: - 30

Pieces: - 11

Solution C o-ordinates: - (1,0) (3,2) (0,0) (2,0) (0,0) (0,0) (0,0) (1,2) (0,0) (0,0) (2,1) (3,0)

Solution: -

Problem

Initial Board

1	2	3	4	5	6
1	2	3	4	5	6
1	2	3	4	5	6

Pieces

Piece 1 Piece 2 Piece 3 Piece 4 Piece 5 Piece 6 Piece 7 Piece 8 Piece 9 Piece 10 Piece 11 Piece 12

Solved successfully. Final board is all 0.

Solution coordinates: (1,0) (3,2) (0,0) (2,0) (0,0) (0,0) (0,0) (1,2) (0,0) (2,1) (3,0)

Solution

Initial Piece 1 (Col1) Piece 2 (Col2) Piece 3 (Col3) Piece 4 (Col4) Piece 5 (Col5)

Problem #5

Depths: - 3
Boards: - 4 * 6
Cells: - 24
Pieces: - 12

Solution Co-ordinates: - (2,0) (3,1) (0,0) (4,0) (2,3) (1,0) (0,1) (3,2) (1,0) (0,0) (2,1) (0,0) (1,0)

Solution: -

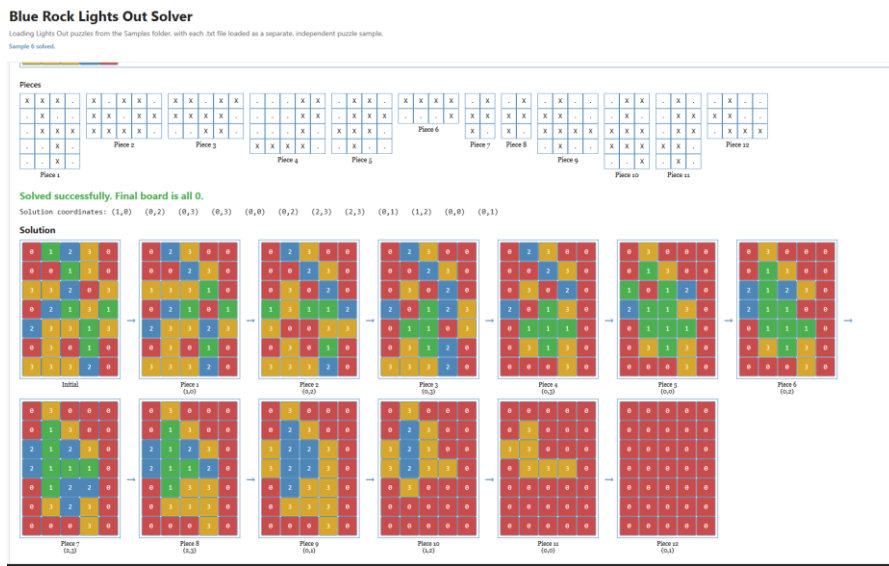


Problem #6

Depths: - 4
Boards: - 7 * 7
Cells: - 35
Pieces: - 12

Solution Co-ordinates: - (1,0) (0,2) (0,3) (0,3) (0,0) (0,2) (2,3) (2,3) (0,1) (1,2) (0,0) (0,1)

Solution: -

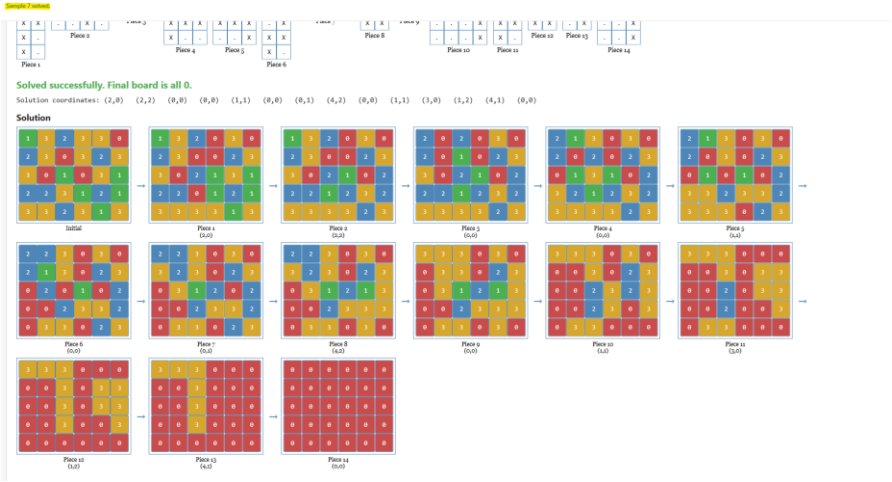


Problem #7

Depths: - 4
Boards: - 5 * 6
Cells: - 30
Pieces: - 14

Solution Co-ordinates : - (1,0) (0,2) (0,3) (0,3) (0,0) (0,2) (2,3) (2,3) (0,1) (1,2) (0,0) (0,1)

Solution: -

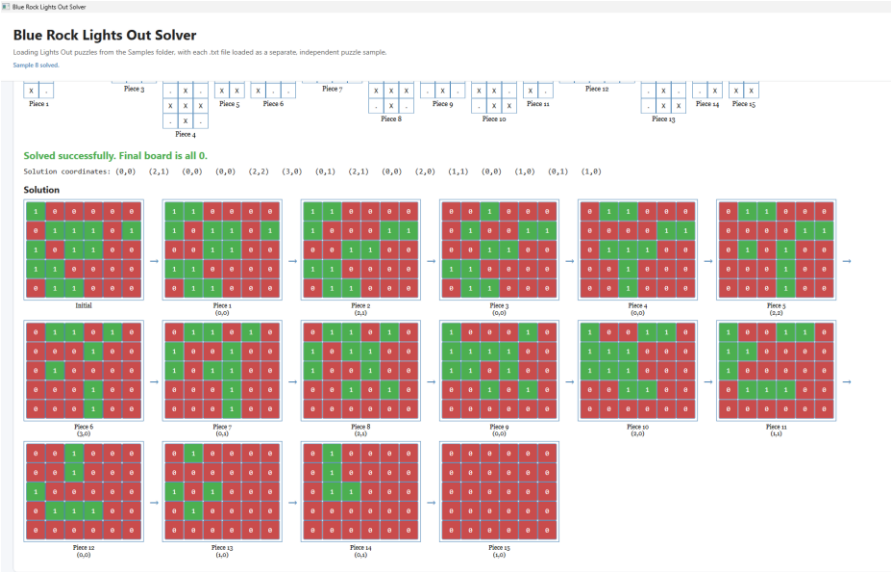


Problem #8

Depths: - 2
Boards: - 5 * 6
Cells: - 30
Pieces: - 15

Solution Co-ordinates: - (0,0) (2,1) (0,0) (0,0) (2,2) (3,0) (0,1) (2,1) (0,0) (2,0) (1,1) (0,0) (1,0) (0,1) (1,0)

Solution: -

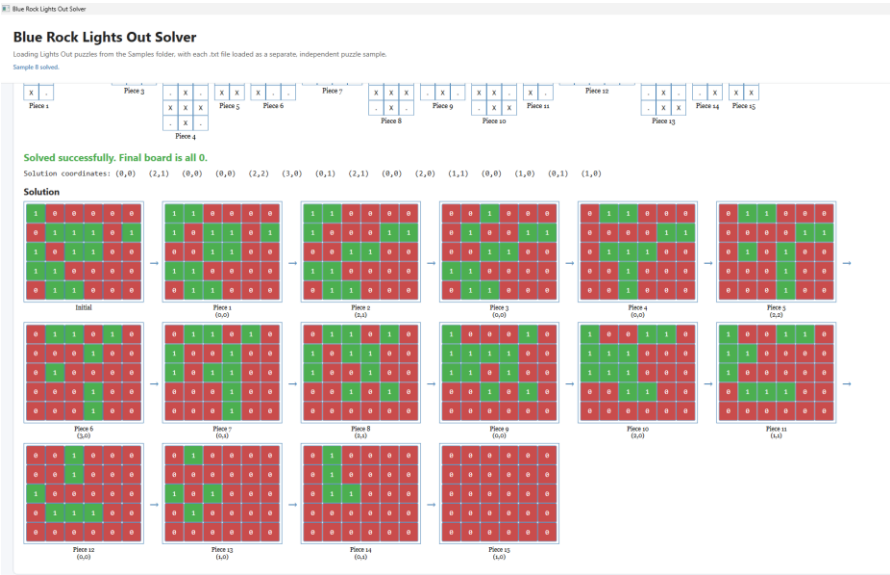


Problem #9

Depths: - 2
Boards: - 6 * 8
Cells: - 48
Pieces: - 16

Solution Co-ordinates : - (1,0) (4,4) (1,1) (1,1) (3,3) (2,0) (2,2) (0,1) (1,0) (5,3) (0,0) (2,0) (4,1) (3,3) (0,3) (2,2)

Solution: -



Problem #10

Depths: - 4
Boards: - 9 * 7
Cells: - 63
Pieces: - 17

Solution Co-ordinates : - (4,2) (2,0) (0,0) (4,2) (0,2) (1,1) (3,3) (4,0) (2,3) (1,3) (1,2) (2,0) (2,0) (1,3) (3,6) (2,2) (0,3)

Solution: -

